

NyumbaAI: Groq AI-Driven PROPERTY FINDER

Revolutionizing Home searching with AI

**Where Technology Meets
Home**

Challenges when looking for a home

Finding a home is stressful, expensive, and time-consuming.



Why NyumbaAI ?

- Find your home in minutes.
- Save money.
- Know where to look.

With NyumbaAI



Key Features

- Natural language property search across Kenya
- Location extraction from user queries
- Property recommendations based on ratings and reviews
- Interactive property listings with detailed information
- Interactive map visualization of property locations
- Search history tracking
- LLM-powered property summaries and recommendations using Groq

How it works

Step-by-Step Visual Journey:

1. 📝 User Prompt:

- User types natural language queries (e.g., "Affordable 2-bedroom apartments in Kilimani").

2. 🧠 AI Understanding:

- GROQ AI analyzes the query, extracts location and property needs.

3. 🔍 Smart Search Engine:

- SerpAPI fetches real-time property listings based on user intent.

4. 🏠 Result Display:

- Streamlit app organizes properties into detailed cards and interactive maps.

5. 🌐 User Experience Enhancements:

- VR tours (*future upgrade*), map view, recommendations, and booking support.



Our Tech Stack

Frontend:

Django— for fast, interactive user interfaces.

Python — for seamless logic and API integration.

Backend & APIs:

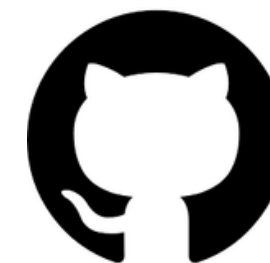
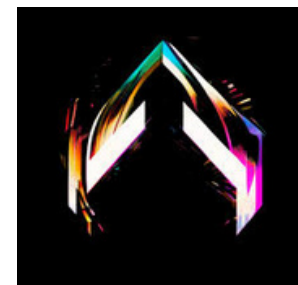
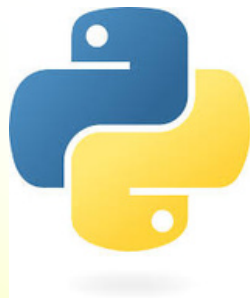
GROQ AI API —for natural language understanding and property Recommendations.

SerpAPI — to fetch real-time search results about properties.

Other Services:

Google Maps API (future) — for mapping properties and nearby amenities.

GitHub — for version control and collaboration.



Demo



Content Structure:

◆ How We Will Showcase:

Step 1: User enters a housing query (e.g., “Affordable apartments in Westlands under KSh 50,000”).

Step 2: NyumbaAI processes the input using GROQ and SerpAPI.

Step 3: Instant property listings appear with details and maps.

Step 4: (Optional) Showcase mockup VR experience or future roadmap hint.

◆ Highlighted Features During Demo:

Fast natural language search.

Real-time property recommendations.

Interactive property map integration.

Clean, intuitive user interface (Streamlit).

Future Vision

- Price filtering
- Collaboration with landlords and agents
- Collaboration with moving companies
- Smart neighborhood insights
- VR house tours

